Mathematical and Computational Methods

Unit 4: Vectors

With this unit the module turns from numbers and functions to *linear algebra*, whose objects are *vectors* and the matrices that act on them. A vector carries a size and a direction, as a displacement, a velocity, a force or a list of measurements does. This unit develops their algebra, the two ways of *multiplying* them—the scalar (dot) and vector (cross) products—and the equations of the *lines* and *planes* they let us write down.

Where this fits. The coordinate plane of Unit 1, Pythagoras and the cosine rule are the only prerequisites; no calculus is needed. Two ideas stay at the level of intuition until Unit 5 makes them computational: testing vectors for independence, and the *determinant* hiding inside the cross product. The lines and planes of Section 3 are solution sets of linear systems, which Unit 5 solves, and the strand closes with eigensystems in Unit 6.

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Learning objectives

By the end of this unit you will be able to:

- Add, subtract and scale vectors, and find magnitudes and unit vectors.
- Compute the dot product (angle, projection, orthogonality) and the cross product (normal, area).
- Derive and use the equations of lines and planes.
- Explain linear independence and basis, and interpret cosine similarity in $\mathbb{R}^n$.

1 Vectors and their algebra

1.1 Scalars and vectors

A *scalar* is a quantity described by a single number—a temperature, a mass, a length. A *vector* needs a number *and* a direction: a wind of 30 km/h *towards the north-east*, or a force of 10 N *downward*. We draw a vector as an arrow whose length is its magnitude and whose heading is its direction.

Definition 1.1 (Vector). Fix coordinate axes. A *vector* is an ordered list of real numbers, its *components*,

$$\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2, \qquad \mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3, \qquad \text{or} \qquad \mathbf{v} = (v_1, \ldots, v_n) \in \mathbb{R}^n.$$

Vectors with the same number of components (the same *dimension*) are added, and a vector is scaled by a real number k, *componentwise*:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \ldots, u_n + v_n), \qquad k\mathbf{v} = (kv_1, \ldots, kv_n).$$

What makes $\mathbb{R}^n$ a space of vectors is exactly that it is *closed* under these two operations: adding two vectors, or scaling one, again gives a vector of the same kind. The *zero vector* $\mathbf{0} = (0, \ldots, 0)$ changes nothing when it is added: $\mathbf{u} + \mathbf{0} = \mathbf{u}$.

We write vectors in bold, $\mathbf{u}, \mathbf{v}, \mathbf{w}$, and scalars in ordinary italic, a, k, t. (By hand, an underline $\underline{v}$ or an arrow $\overrightarrow{v}$ is usual.)

Note (Geometric reading). In $\mathbb{R}^2$ and $\mathbb{R}^3$ a vector is pictured as a *directed segment* (an arrow) from an *initial point* to a *terminal point*. Two arrows with the same length and direction—one a parallel translate of the other—represent the *same* vector; sliding $\mathbf{v}$ so its tail is at the origin, the coordinates of its tip are the components above.

Note (Intuition, not the definition). It is often helpful to think of a nonzero vector as carrying a *magnitude* and a *direction*; but that picture is only a label for the data above, and it is the two operations—addition and scaling—not the words "magnitude" and "direction"—that make something a vector. (The abstract notion of a *vector space*, of which $\mathbb{R}^n$ is the first example, is developed in the separate, later Linear Algebra module taken by the Mathematics and Data Science students.)

1.2 Components

Conversely, any two points determine a vector.

Definition 1.2 (The vector between two points). The vector from a point $P(p_1, p_2, p_3)$ to a point $Q(q_1, q_2, q_3)$ is found by subtracting coordinates,

$$\overrightarrow{PQ} = (q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3),$$

with the obvious two-term version in $\mathbb{R}^2$.

Note (The standard basis). Writing $\mathbf{i} = (1,0,0)$, $\mathbf{j} = (0,1,0)$, $\mathbf{k} = (0,0,1)$ for the unit vectors along the axes, every vector in $\mathbb{R}^3$ is built from them in exactly one way:

$$\mathbf{v} = (v_1, v_2, v_3) = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}.$$

The two notations—triples and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ combinations—are interchangeable; in $\mathbb{R}^2$ one drops the third entry and uses $\mathbf{i}, \mathbf{j}$.

1.3 Addition, subtraction and scalar multiplication

Addition and scalar multiplication were defined componentwise in the definition of a vector; each has a clean geometric meaning, and *subtraction* is the one further operation we need.

Definition 1.3 (Subtraction). For $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$, subtraction is addition of the negative,

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-1)\mathbf{v} = (u_1 - v_1,\ u_2 - v_2,\ u_3 - v_3).$$

Geometrically, $\mathbf{u} + \mathbf{v}$ is found by the *triangle rule*—place the tail of $\mathbf{v}$ at the tip of $\mathbf{u}$ and join start to finish—equivalently the diagonal of the parallelogram on $\mathbf{u}$ and $\mathbf{v}$ (Figure 1). Scalar multiplication $k\mathbf{v}$ stretches $\mathbf{v}$ by the factor $|k|$, keeping its direction if $k > 0$ and reversing it if $k < 0$.

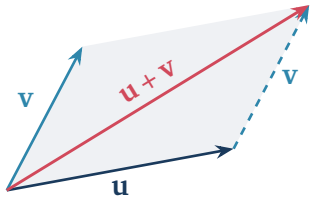


Figure 1: The triangle (and parallelogram) rule for $\mathbf{u} + \mathbf{v}$.

Example 1.1 ***(Component arithmetic).*** With $\mathbf{u} = (1,3)$ and $\mathbf{v} = (4,-1)$,

$$\mathbf{u} + \mathbf{v} = (5,2), \qquad \mathbf{u} - \mathbf{v} = (-3,4), \qquad 2\mathbf{u} = (2,6).$$

These operations obey the rules one expects, proved in each case by checking one component at a time.

Proposition 1.1 (Algebraic properties). For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars a, k,

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}, \quad (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w}), \quad \mathbf{u} + \mathbf{0} = \mathbf{u}, \quad \mathbf{u} + (-\mathbf{u}) = \mathbf{0},$$

$$k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}, \quad (a + k)\mathbf{v} = a\mathbf{v} + k\mathbf{v}, \quad a(k\mathbf{v}) = (ak)\mathbf{v}, \quad 1\,\mathbf{v} = \mathbf{v}.$$

1.4 Magnitude and unit vectors

The length of a vector is read straight off its components by Pythagoras (Figure 2).

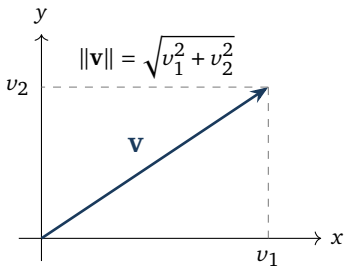


Figure 2: A vector and its components; the magnitude is the hypotenuse.

Definition 1.4 (Magnitude). The *magnitude* (or *norm*, or *length*) of $\mathbf{v} = (v_1, v_2, v_3)$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2},$$

with the obvious two-term version in $\mathbb{R}^2$. The distance between points P and Q is $\left\|\overrightarrow{PQ}\right\|$.

A vector of length one is a *unit vector*; it records a pure direction. Dividing any nonzero vector by its own length produces the unit vector pointing the same way, a step called *normalising*.

Definition 1.5 (Unit vector). For $\mathbf{v} \neq \mathbf{0}$, the unit vector in the direction of $\mathbf{v}$ is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Example 1.2 *(Normalising).* For $\mathbf{v} = (2, -1, 2)$ the length is $\|\mathbf{v}\| = \sqrt{4 + 1 + 4} = 3$, so the unit vector in its direction is $\hat{\mathbf{v}} = \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right)$. As a check, $\|\hat{\mathbf{v}}\| = \frac{1}{3}\sqrt{4 + 1 + 4} = 1$.

1.5 Beyond three dimensions: $\mathbb{R}^n$ and a data-science note

Nothing in the component definitions used the number three. A list of n real numbers

$$\mathbf{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n, \qquad \|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2},$$

is a vector in *n-space*, added, scaled and measured by the same formulas as before. We cannot *draw* $\mathbb{R}^n$ once n exceeds three, but the algebra is unchanged, and that is exactly what data science needs.

Remark (Data points are vectors). A record with n measured features—the pixels of an image, the word counts of a document, the readings of a sensor array—is a point of $\mathbb{R}^n$, hence a vector. Their *directions* often matter more than their lengths (a document and its doubled copy should count as identical in subject), which is why the angle between two such vectors, introduced through the dot product in Section 2, becomes the standard *cosine-similarity* measure. The geometry of this unit is therefore not confined to two and three dimensions.

1.6 Linear combinations, independence and basis (intuition)

Adding scaled vectors, $a\mathbf{u} + b\mathbf{v} + c\mathbf{w}$, gives a *linear combination* of them. Two questions recur throughout linear algebra: is a vector *redundant*—a combination of the others—and can a given set *reach every vector*?

Note (Independence and basis, informally). A set of vectors is *linearly independent* if none of them is a linear combination of the rest—there is no redundancy. (Equivalently, and in the form the later Linear Algebra module uses: the only linear combination that equals $\mathbf{0}$ is the one with all coefficients zero.) A *basis* is an independent set large enough to build every vector in the space, each in exactly one way. In $\mathbb{R}^3$ the standard $\mathbf{i}, \mathbf{j}, \mathbf{k}$ form the basic example: two vectors are independent unless they are parallel (counting $\mathbf{0}$ as parallel to everything), and three are independent unless they lie in a common plane. In $\mathbb{R}^2$ two independent vectors already reach every vector, so three vectors in $\mathbb{R}^2$ are never independent: one of them is always a combination of the others. The systematic *test* for independence—and the general procedure that finds the coefficients of a combination—is the solution of a linear system by *Gaussian elimination*, which we develop in Unit 5. For now the geometric picture suffices.

In $\mathbb{R}^2$ the coefficients can already be found by hand, by matching components. To write $(2, 5)$ as $\alpha(4, 1) + \beta(1, -2)$, compare first and then second components:

$$4\alpha + \beta = 2, \qquad \alpha - 2\beta = 5.$$

The first gives $\beta = 2 - 4\alpha$; substituting, $\alpha - 2(2 - 4\alpha) = 5$, so $9\alpha = 9$, giving $\alpha = 1$ and $\beta = -2$. Check: $(4, 1) - 2(1, -2) = (2, 5)$. Because $(4, 1)$ and $(1, -2)$ are not parallel, the same matching works for every target vector, so they form a basis of $\mathbb{R}^2$. By contrast $(4, 1)$ and $(8, 2)$ are parallel, hence dependent, and their combinations reach only the line through the origin that they share.

2 Products of vectors

There is no single "multiplication" of vectors. Two genuinely different products are useful: one returns a *scalar* and measures alignment; the other (in three dimensions) returns a *vector* perpendicular to its factors.

2.1 The scalar (dot) product

Definition 2.1 (Dot product). The *dot product* of $\mathbf{u}$ and $\mathbf{v}$ is the scalar

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 \qquad (\text{and } u_1v_1 + \cdots + u_nv_n \text{ in } \mathbb{R}^n).$$

The same number has a purely geometric description, and the link between the two is the whole reason the dot product is useful.

Theorem 2.1 (Geometric form of the dot product). If $\theta \in [0, \pi]$ is the angle between nonzero $\mathbf{u}$ and $\mathbf{v}$, then

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta.$$

Proof. Place $\mathbf{u}$ and $\mathbf{v}$ tail to tail; the third side of the triangle is $\mathbf{u} - \mathbf{v}$. The law of cosines gives $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\,\|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta$. But expanding in components, $\|\mathbf{u} - \mathbf{v}\|^2 = \sum_i (u_i - v_i)^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\sum_i u_iv_i$. Comparing the two expressions leaves $\sum_i u_iv_i = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta$. ∎

The dot product is commutative, distributes over addition, pulls scalars out of either factor, and a vector dotted with itself returns its length squared:

$$\mathbf{u}\cdot\mathbf{v}=\mathbf{v}\cdot\mathbf{u},\qquad \mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w},\qquad (k\mathbf{u})\cdot\mathbf{v}=k(\mathbf{u}\cdot\mathbf{v}),\qquad \mathbf{v}\cdot\mathbf{v}=\|\mathbf{v}\|^2.$$

The angle between two vectors

Solving the geometric form for the cosine gives the formula that drives everything else.

Note (Angle and its sign).

$$\cos\theta=\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.$$

Here $\mathbf{u}$ and $\mathbf{v}$ are both nonzero, so the angle $\theta\in[0,\pi]$ is defined. The sign of $\mathbf{u}\cdot\mathbf{v}$ alone reports the angle's type: positive means *acute*, negative *obtuse*, and zero a *right angle*. Parallel vectors sit at the two ends of $[0,\pi]$: $\theta=0$ if they point the same way ($\cos\theta=1$), and $\theta=\pi$ if they point in opposite directions ($\cos\theta=-1$).

Example 2.1 ***(Angle between two vectors).*** Find the angle between $\mathbf{u}=(1,0,1)$ and $\mathbf{v}=(1,1,0)$. The ingredients are

$$\mathbf{u}\cdot\mathbf{v}=1\cdot1+0\cdot1+1\cdot0=1,\qquad \|\mathbf{u}\|=\sqrt{2},\qquad \|\mathbf{v}\|=\sqrt{2},$$

so

$$\cos\theta=\frac{1}{\sqrt{2}\,\sqrt{2}}=\frac{1}{2}\quad\Longrightarrow\quad\theta=\frac{\pi}{3}=60^\circ.$$

The same three steps—dot the vectors, divide by the lengths, take the inverse cosine—give the angle in any dimension, and in $\mathbb{R}^n$ this cosine is the *cosine similarity* of Section 1.

Orthogonality and projection

Definition 2.2 (Orthogonality test). Vectors $\mathbf{u}$ and $\mathbf{v}$ are *orthogonal* (perpendicular) precisely when

$$\mathbf{u}\cdot\mathbf{v}=0.$$

The zero vector is taken to be orthogonal to everything.

The dot product also resolves one vector along another. The *projection* of $\mathbf{u}$ onto $\mathbf{v}$ is the shadow $\mathbf{u}$ casts on the line through $\mathbf{v}$ (Figure 3); its signed length is the *scalar projection* $\|\mathbf{u}\|\cos\theta=\dfrac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}$.

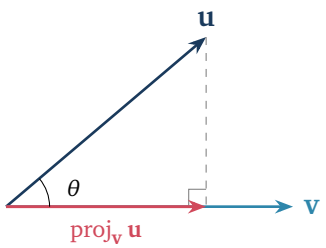


Figure 3: The vector projection of $\mathbf{u}$ onto $\mathbf{v}$.

Definition 2.3 (Vector projection). The (vector) projection of $\mathbf{u}$ onto a nonzero $\mathbf{v}$ is

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\,\mathbf{v} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\,\mathbf{v}.$$

Example 2.2 *(A projection).* Project $\mathbf{u} = (4,3)$ onto $\mathbf{v} = (1,2)$. Here $\mathbf{u}\cdot\mathbf{v} = 4+6 = 10$ and $\|\mathbf{v}\|^2 = 1+4 = 5$, so

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{10}{5}\,(1,2) = (2,4).$$

This is $\mathbf{u}_{\parallel}$, and the remainder $\mathbf{u}_{\perp} = \mathbf{u} - \mathbf{u}_{\parallel} = (2,-1)$ is perpendicular to $\mathbf{v}$: $\mathbf{u}_{\perp}\cdot\mathbf{v} = 2-2 = 0$.

Remark (Physics: work). When a constant force $\mathbf{F}$ moves an object through a displacement $\mathbf{d}$, only the component of the force *along* the motion does work, and the work done is exactly the dot product $W = \mathbf{F}\cdot\mathbf{d} = \|\mathbf{F}\|\,\|\mathbf{d}\|\cos\theta$, in joules ($1\,\mathrm{J} = 1\,\mathrm{N\,m}$). That component, the scalar projection of $\mathbf{F}$ onto $\mathbf{d}$, is $\mathbf{F}\cdot\hat{\mathbf{d}} = (\mathbf{F}\cdot\mathbf{d})/\|\mathbf{d}\| = W/\|\mathbf{d}\|$. A force perpendicular to the motion ($\theta = 90^\circ$) does no work—the orthogonality test in physical dress.

2.2 The vector (cross) product

In $\mathbb{R}^3$, and only there, a second product returns a *vector* perpendicular to both factors.

Definition 2.4 (Cross product). For $\mathbf{u} = (u_1,u_2,u_3)$ and $\mathbf{v} = (v_1,v_2,v_3)$ in $\mathbb{R}^3$,

$$\mathbf{u}\times\mathbf{v} = \big(u_2v_3 - u_3v_2,\;\; u_3v_1 - u_1v_3,\;\; u_1v_2 - u_2v_1\big).$$

The three components are awkward to memorise directly, but they are exactly the pattern of a 3×3 *determinant* with the basis vectors in the top row.

Note (The determinant mnemonic).

$$\mathbf{u}\times\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \mathbf{i}\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \mathbf{j}\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \mathbf{k}\begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix},$$

where a 2×2 determinant is $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$; it is a memory aid until Unit 5.

Theorem 2.2 (Properties of the cross product). For $\mathbf{u},\mathbf{v},\mathbf{w}\in\mathbb{R}^3$ and a scalar k:

1. $\mathbf{u}\times\mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$: $\quad\mathbf{u}\cdot(\mathbf{u}\times\mathbf{v}) = 0$ and $\mathbf{v}\cdot(\mathbf{u}\times\mathbf{v}) = 0$.
2. It is *anticommutative*: $\mathbf{v}\times\mathbf{u} = -(\mathbf{u}\times\mathbf{v})$; in particular $\mathbf{u}\times\mathbf{u} = \mathbf{0}$.
3. $\|\mathbf{u}\times\mathbf{v}\| = \|\mathbf{u}\|\,\|\mathbf{v}\|\sin\theta$, where θ is the angle between nonzero $\mathbf{u}$ and $\mathbf{v}$: the area of the parallelogram with sides $\mathbf{u}$ and $\mathbf{v}$.
4. $\mathbf{u}\times\mathbf{v} = \mathbf{0}$ exactly when $\mathbf{u}$ and $\mathbf{v}$ are parallel (with the zero vector taken to be parallel to everything, covering the case where a factor is $\mathbf{0}$ and θ is undefined).

5. It distributes over addition, $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$, and $(k\mathbf{u}) \times \mathbf{v} = k(\mathbf{u} \times \mathbf{v})$.

Two opposite directions are perpendicular to both $\mathbf{u}$ and $\mathbf{v}$; the *right-hand rule* says which one $\mathbf{u} \times \mathbf{v}$ takes: with the fingers curling from $\mathbf{u}$ to $\mathbf{v}$, the thumb points along $\mathbf{u} \times \mathbf{v}$ (Figure 4). For the basis vectors this gives the cycle $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$, and the reverse order flips each sign. Because the magnitude is $\|\mathbf{u}\| \, \|\mathbf{v}\| \sin\theta$, the cross product measures *turning* and *area* just as the dot product measures *alignment*.

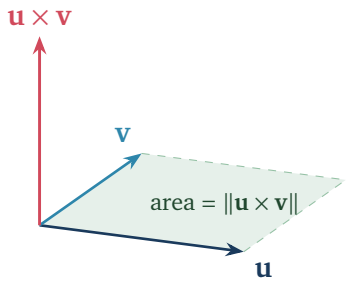


Figure 4: $\mathbf{u} \times \mathbf{v}$ is perpendicular to the plane of $\mathbf{u}, \mathbf{v}$; its length is the parallelogram's area. (Schematic: $\mathbf{u} \times \mathbf{v}$ points out of the plane of $\mathbf{u}, \mathbf{v}$, not within the page.)

Example 2.3 ***(A cross product, with checks).*** Let $\mathbf{u} = (2, 1, -1)$ and $\mathbf{v} = (1, 3, 2)$. Then

$$\mathbf{u} \times \mathbf{v} = \big(1 \cdot 2 - (-1) \cdot 3, \;\; (-1) \cdot 1 - 2 \cdot 2, \;\; 2 \cdot 3 - 1 \cdot 1\big) = (5, -5, 5).$$

The orthogonality property checks out: $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 10 - 5 - 5 = 0$ and $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 5 - 15 + 10 = 0$. Its length is $\|\mathbf{u} \times \mathbf{v}\| = \sqrt{25 + 25 + 25} = 5\sqrt{3}$, so the triangle with sides $\mathbf{u}$ and $\mathbf{v}$ has area $\frac{1}{2} \cdot 5\sqrt{3} = \frac{5\sqrt{3}}{2}$.

Remark (Physics: torque and rotation). The cross product is the natural language of rotation. The *torque* of a force $\mathbf{F}$ applied at position $\mathbf{r}$ from a pivot is $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$, and the magnetic part of the *Lorentz force* on a charge q moving with velocity $\mathbf{v}$ in a field $\mathbf{B}$ is $q\,\mathbf{v} \times \mathbf{B}$. In each case the output is perpendicular to the inputs, with a magnitude that vanishes when they are parallel—the geometry of properties 1 and 3 made physical.

Optional: the scalar triple product

Remark (Volume and coplanarity (enrichment)). Combining the two products, the *scalar triple product* $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is a single number, equal to the 3×3 determinant whose rows are the three vectors:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.$$

Its absolute value is the *volume* of the slanted box (a *parallelepiped*) built on $\mathbf{u}, \mathbf{v}, \mathbf{w}$, so the triple product is *zero* exactly when the three vectors are *coplanar*: a test for three-vector independence. Reordering the three vectors changes the triple product at most in sign (a cyclic reordering, such as $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{u})$, leaves it unchanged), so the volume does not depend

on the order. This is also the determinant of Unit 5 making an early appearance: a zero determinant will mean precisely a flattened, non-invertible transformation.

Example 2.4 ***(A volume, and a coplanarity test).*** The parallelepiped built on $\mathbf{u} = (1, 2, 0)$, $\mathbf{v} = (0, 3, 1)$ and $\mathbf{w} = (2, 0, 1)$ has volume $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$. First

$$\mathbf{v} \times \mathbf{w} = \big(3 \cdot 1 - 1 \cdot 0,\ \ 1 \cdot 2 - 0 \cdot 1,\ \ 0 \cdot 0 - 3 \cdot 2\big) = (3, 2, -6),$$

and then $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = 1 \cdot 3 + 2 \cdot 2 + 0 \cdot (-6) = 7$, so the volume is 7. Equivalently, expanding the 3×3 determinant whose rows are the three vectors along its first row,

$$\begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 0 & 1 \end{vmatrix} = 1 \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} + 0 \begin{vmatrix} 0 & 3 \\ 2 & 0 \end{vmatrix} = 1(3) - 2(-2) + 0 = 7,$$

the same value by the determinant form promised above. For the coplanarity test, replace $\mathbf{u}$ by $\mathbf{u}' = \mathbf{v} + \mathbf{w} = (2, 3, 2)$, which lies in the plane of $\mathbf{v}$ and $\mathbf{w}$ by construction. As expected its triple product vanishes,

$$\mathbf{u}' \cdot (\mathbf{v} \times \mathbf{w}) = 2 \cdot 3 + 3 \cdot 2 + 2 \cdot (-6) = 0,$$

the zero signalling a collapsed, zero-volume box—exactly the three-vector dependence flagged in Section 1.

3 Lines and planes

A point fixes *where*; a vector fixes *which way*. Together they pin down the two basic flat objects of space—the line and the plane—and the products of Section 2 turn those descriptions into equations.

3.1 The equation of a line

A line is determined by one point on it and a direction to travel along. Starting from the position vector $\mathbf{r}_0$ of a known point and stepping any multiple of a direction vector $\mathbf{d}$ sweeps out the whole line (Figure 5).

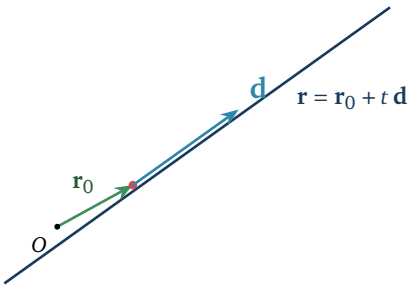


Figure 5: A line as a point $\mathbf{r}_0$ plus all multiples of a direction $\mathbf{d}$.

Definition 3.1 (Vector and parametric equations of a line). The line through the point with position vector $\mathbf{r}_0$ in the direction of a nonzero $\mathbf{d} = (d_1, d_2, d_3)$ is

$$\mathbf{r} = \mathbf{r}_0 + t\,\mathbf{d}, \qquad t \in \mathbb{R} \qquad \text{(vector form)},$$

which in coordinates is the *parametric form*

$$x = x_0 + t\,d_1, \qquad y = y_0 + t\,d_2, \qquad z = z_0 + t\,d_3.$$

Eliminating t (when no d_i is zero) gives the *symmetric form*

$$\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2} = \frac{z - z_0}{d_3}.$$

In the plane $\mathbb{R}^2$ the same equation, with two components, is just the familiar straight line: when $d_1 \neq 0$, writing t in terms of x recovers $y = mx + c$ with slope $m = d_2/d_1$; when $d_1 = 0$ the line is the vertical $x = x_0$, which no $y = mx + c$ can express.

Example 3.1 ***(Line through two points).*** The line through $A(1, 2, -1)$ and $B(3, -1, 2)$ has direction $\mathbf{d} = \overrightarrow{AB} = (2, -3, 3)$. Taking A as the base point,

$$\mathbf{r} = (1, 2, -1) + t\,(2, -3, 3), \qquad \text{i.e.} \quad x = 1 + 2t, \;\; y = 2 - 3t, \;\; z = -1 + 3t,$$

and in symmetric form $\dfrac{x-1}{2} = \dfrac{y-2}{-3} = \dfrac{z+1}{3}$. Putting $t = 1$ returns the point B, as it should.

3.2 The equation of a plane

A line needed a direction to follow; a plane is pinned down instead by the one direction it must *avoid*—a *normal* vector $\mathbf{n}$ perpendicular to every line in the plane (Figure 6). A point $\mathbf{r}$ lies in the plane through $\mathbf{r}_0$ exactly when the displacement $\mathbf{r} - \mathbf{r}_0$ lies flat, i.e. is orthogonal to $\mathbf{n}$—and orthogonality is a dot product set to zero.

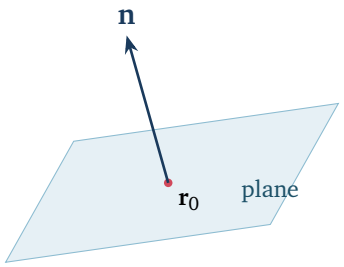


Figure 6: A plane is fixed by a point $\mathbf{r}_0$ and a normal $\mathbf{n}$.

Definition 3.2 (Equations of a plane in $\mathbb{R}^3$). The plane through the point with position vector $\mathbf{r}_0 = (x_0, y_0, z_0)$ with nonzero normal $\mathbf{n} = (a, b, c)$ is, in *normal (point-normal) form*,

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0,$$

which multiplies out to the *scalar (Cartesian) form*

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0, \qquad \text{i.e.} \quad ax + by + cz = d,$$

with $d = \mathbf{n} \cdot \mathbf{r}_0$. Alternatively, given two non-parallel vectors $\mathbf{u}, \mathbf{v}$ lying in the plane, the *parametric form* is

$$\mathbf{r} = \mathbf{r}_0 + s\,\mathbf{u} + t\,\mathbf{v}, \qquad s, t \in \mathbb{R}.$$

The coefficients in the Cartesian equation are exactly the components of a normal: the plane $ax + by + cz = d$ has normal $\mathbf{n} = (a, b, c)$, read off by inspection.

Example 3.2 ***(The plane through three points).*** Find the plane through $P(1, 0, 2)$, $Q(2, 1, 0)$ and $R(0, 2, 1)$.

Two vectors in the plane. Using P as the base point,

$$\overrightarrow{PQ} = (1, 1, -2), \qquad \overrightarrow{PR} = (-1, 2, -1).$$

A normal, by the cross product. A vector perpendicular to both lies normal to the plane:

$$\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \big(1 \cdot (-1) - (-2) \cdot 2,\ \ (-2) \cdot (-1) - 1 \cdot (-1),\ \ 1 \cdot 2 - 1 \cdot (-1)\big) = (3, 3, 3).$$

Any nonzero scalar multiple serves, so take the tidier $\mathbf{n} = (1, 1, 1)$.

The equation. Point-normal form with the point P gives

$$1(x - 1) + 1(y - 0) + 1(z - 2) = 0 \quad \Longrightarrow \quad x + y + z = 3.$$

As a check, each of P, Q, R satisfies it: $1 + 0 + 2 = 2 + 1 + 0 = 0 + 2 + 1 = 3$. The same two vectors give the parametric form, $\mathbf{r} = (1, 0, 2) + s\,(1, 1, -2) + t\,(-1, 2, -1)$. This single example uses every tool of the unit—subtraction of points, the cross product for a normal, and the dot product hidden in the point-normal form.

3.3 Where a line meets a plane; the angle between two planes

With a line and a plane both in hand, two natural questions are pure dot-product geometry: where a line crosses a plane, and at what angle two planes meet. Each reuses the very objects built above.

Example 3.3 ***(A line meeting a plane).*** Where does the line $\mathbf{r} = (1, 2, -1) + t\,(2, -3, 3)$ of the earlier example meet the plane $x + y + z = 3$? A point of the line has coordinates $x = 1 + 2t$, $y = 2 - 3t$, $z = -1 + 3t$; substituting these into the plane equation,

$$(1 + 2t) + (2 - 3t) + (-1 + 3t) = 3 \quad \Longrightarrow \quad 2 + 2t = 3 \quad \Longrightarrow \quad t = \tfrac{1}{2}.$$

The single parameter value locates the crossing point: putting $t = \frac{1}{2}$ back into the line gives $\big(2, \frac{1}{2}, \frac{1}{2}\big)$, and as a check $2 + \frac{1}{2} + \frac{1}{2} = 3$.

Note (When does a line cross a plane?). The crossing above existed because the line's direction $\mathbf{d} = (2, -3, 3)$ was *not* orthogonal to the plane's normal $\mathbf{n} = (1, 1, 1)$: here $\mathbf{d}\cdot\mathbf{n} = 2 - 3 + 3 = 2 \neq 0$, so the substitution left a genuine equation for t with one root. When instead $\mathbf{d} \cdot \mathbf{n} = 0$ the direction lies flat in the plane, and the line is either *parallel* to it (the equation for t has no solution) or *contained* in it (every t works). The dot product cannot tell those two apart—it is 0 either way—so the *base point* decides: the line lies in the plane exactly when $\mathbf{r}_0$ satisfies the plane equation, and is parallel to it otherwise.

Example 3.4 ***(The angle between two planes).*** The angle between two planes is the angle between their normals: two planes are tilted apart exactly as much as the directions they forbid. Reversing one normal would turn that angle θ into $\pi - \theta$, so we report the smaller one, $\theta \in [0, \pi/2]$, by taking the absolute value of the dot product. For the plane $x + y + z = 3$, with normal $\mathbf{n}_1 = (1, 1, 1)$, and the plane $x - y + 2z = 0$, with normal $\mathbf{n}_2 = (1, -1, 2)$,

$$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \, \|\mathbf{n}_2\|} = \frac{1 - 1 + 2}{\sqrt{3}\,\sqrt{6}} = \frac{2}{\sqrt{18}} = \frac{\sqrt{2}}{3} \approx 0.471,$$

so $\theta \approx 61.9°$: the dot product of Section 2, now measuring two flat sheets.

Note (Parallel planes and their intersection). Two distinct planes either are *parallel*—when their normals are parallel, as for $x + y + z = 3$ and $2x + 2y + 2z = 1$—or else meet in a *line*, the case just computed; perpendicular planes, with $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$, meet at $\theta = \pi/2$. The line of intersection solves the two equations together: two equations in three unknowns, so one unknown is left free as the line's parameter.

Example 3.5 ***(Distance from a point to a plane).*** How far is the origin from the plane $x + y + z = 3$? Step from O along the normal direction $\mathbf{n} = (1, 1, 1)$: the point $t\,\mathbf{n} = (t, t, t)$ reaches the plane when $3t = 3$, i.e. at $t = 1$, so the foot of the perpendicular is $(1, 1, 1)$ and the distance is $\|(1, 1, 1)\| = \sqrt{3}$. The same steps work for any point $\mathbf{p} = (p_1, p_2, p_3)$ and any plane $ax + by + cz = d$: the point $\mathbf{p} + t\,\mathbf{n}$ lies on the plane when $\mathbf{n} \cdot \mathbf{p} + t\,\|\mathbf{n}\|^2 = d$, so $t = (d - \mathbf{n} \cdot \mathbf{p})/\|\mathbf{n}\|^2$, and the distance $\|t\,\mathbf{n}\| = |t| \, \|\mathbf{n}\|$ is

$$\frac{|ap_1 + bp_2 + cp_3 - d|}{\|\mathbf{n}\|} = \frac{|0 + 0 + 0 - 3|}{\sqrt{3}} = \sqrt{3},$$

as found above.

Summary

Concept	Key idea
Vector	Ordered list $\mathbf{v} = (v_1, \ldots, v_n) \in \mathbb{R}^n$, closed under componentwise + and scalar $\times$; pictured as an arrow that may be slid anywhere (magnitude and direction).
Operations	Componentwise $\mathbf{u} \pm \mathbf{v}$ and $k\mathbf{v}$; triangle/parallelogram rule; $k\mathbf{v}$ scales length by $\lvert k \rvert$.
Independence, basis	Independent: no vector is a combination of the others (two: not parallel). A basis is independent and builds every vector, each in exactly one way.
Magnitude	$\lVert\mathbf{v}\rVert = \sqrt{v_1^2 + \cdots + v_n^2}$; unit vector $\hat{\mathbf{v}} = \mathbf{v}/\lVert\mathbf{v}\rVert$ (normalising).
Dot product	$\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i = \lVert\mathbf{u}\rVert\, \lVert\mathbf{v}\rVert \cos\theta$; a scalar measuring alignment.
Angle	$\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\lVert\mathbf{u}\rVert\, \lVert\mathbf{v}\rVert}$ (cosine similarity in $\mathbb{R}^n$); sign gives acute/right/obtuse.
Orthogonality	$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$.
Projection	$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\lVert\mathbf{v}\rVert^2}\, \mathbf{v}$.
Cross product	$\mathbf{u} \times \mathbf{v}$ (in $\mathbb{R}^3$): perpendicular to both, anticommutative, $\lVert\mathbf{u} \times \mathbf{v}\rVert = \lVert\mathbf{u}\rVert\, \lVert\mathbf{v}\rVert \sin\theta$ = parallelogram area; $\mathbf{0}$ iff parallel. Determinant mnemonic with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ on top.
Triple product (optional)	$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ = a 3×3 determinant; $\lvert\cdot\rvert$ is parallelepiped volume; zero iff coplanar.
Line	$\mathbf{r} = \mathbf{r}_0 + t\mathbf{d}$ (point + direction); parametric and symmetric forms.
Plane	$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$, i.e. $ax + by + cz = d$ with normal $\mathbf{n} = (a, b, c)$; or $\mathbf{r} = \mathbf{r}_0 + s\mathbf{u} + t\mathbf{v}$. Through three non-collinear points: $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$.
Line meets plane	Substitute the line into the plane and solve for t; if $\mathbf{d} \cdot \mathbf{n} = 0$ the line is parallel to the plane or lies in it.
Planes, distance	Angle between planes from $\lvert\mathbf{n}_1 \cdot \mathbf{n}_2\rvert / (\lVert\mathbf{n}_1\rVert\, \lVert\mathbf{n}_2\rVert)$; distance from $\mathbf{p}$ to $ax + by + cz = d$ is $\lvert\mathbf{n} \cdot \mathbf{p} - d\rvert / \lVert\mathbf{n}\rVert$.

Next unit: matrices—the arrays that store and transform vectors. We build matrix algebra and multiplication, the determinant (promised here) and the inverse, and then use Gaussian elimination to solve the linear systems whose geometry closed this unit.